Used Car Resale Price Prediction

End-to-End Supervised Learning Pipeline using Multiple Linear Regression

AI101 MINI-PROJECT · FINAL EVALUATION

Supervised regression evaluation on structured automotive market data using Scikit-Learn Pipeline, ColumnTransformer, and empirical diagnostic analysis.

PROJECT REPOSITORY & CODEBASE
GitHub: https://github.com/2026aib1001-AI/ai101-mini-project

Group 8

PROJECT TEAM & ROLES

Team Member	Project Role
Siddharth Kumar	Execution Lead
Divyaans Agarwal	Video Lead
A Damaruk Vajra Tej	PPT Lead
Nihal Ahirwar	Documentation Lead

Predicting Used-Car Resale Prices from Multidimensional Vehicle Attributes

PROBLEM & RELEVANCE

Problem Statement

Second-hand vehicle markets suffer from significant valuation opacity, non-linear depreciation dynamics, and severe price dispersion. Estimating fair resale values requires objectively modeling associations across mechanical usage (odometer mileage), vehicle age, mechanical specifications, and manufacturer brand equity.

Market Relevance

Provides prospective car buyers, private sellers, and digital resale platforms with an empirical, data-driven pricing benchmark grounded in verified transaction records rather than arbitrary dealer markups.

Supervised Learning Target: Continuous price prediction (selling_price in ₹) modeled as a multivariate linear function of vehicle usage, aging (fixed 2020 vintage), mechanical specifications, and manufacturer brand equity.

DATASET SPECIFICATION

CarDekho Vehicle Dataset

Dataset Source: CAR DETAILS FROM CAR DEKHO.csv via Kaggle (nehalbirla/vehicle-dataset-from-cardekho)

Component	Attribute Details & Statistics
Dataset Volume	4,340 raw listings → 3,577 clean unique records (763 duplicates removed, 17.6%)
Target Variable (y)	selling_price (Continuous, Range: ₹20,000 to ₹89,00,000; Mean: ₹5.04 Lakhs)
Input Predictors (X)	7 features: age (fixed 2020 vintage: 2020-yr), km_driven, fuel, seller, trans, owner, brand (43 encoded)

Data Quality & Vintage: Vehicle age is derived using the fixed 2020 dataset collection year (range 0–28 yrs). High-cardinality vehicle 'name' (1,491 titles) was dropped to prevent overfitting, retaining brand equity across 27 distinct marques.

Methodology: End-to-End Modular Machine Learning Pipeline

1. Data Ingestion

Load 4,340 CarDekho records from CSV with 8 initial attributes.

2. Data Cleaning

Deduplication: removed 763 duplicate rows (17.6%) → 3,577 clean unique cars.

3. Feature Engineering

Derived car age using fixed ref year (2020 - year) & extracted brand token.

4. Train-Test Split

80% Training (2,861 cars) / 20% Held-Out Testing (716 cars), seed=42.

5. ColumnTransformer

Median + StandardScaler (num); Mode + OneHotEncoder (cat; 43 dims).

6. Model Pipeline

OLS LinearRegression() fitted strictly on train fold; evaluated on held-out test.

FEATURE FORMULATION & PREPROCESSING

Vehicle Age = 2020 - year (fixed ref year) | Brand = name.split()[0]

- **Fixed Reference Year:** 2020 (dataset collection vintage; derived age ranges from 0 to 28 years).
- **Numeric Pipeline:** SimpleImputer(median) + StandardScaler() applied to ['age', 'km_driven'].
- **Categorical Pipeline:** SimpleImputer(most_frequent) + OneHotEncoder(ignore) on ['fuel', 'seller_type', 'transmission', 'owner', 'brand'].
- **Leakage Prevention:** Transformers fitted strictly on X_train (2,861); transformed X_test (716) without lookahead.

PIPELINE SPECIFICATIONS & MODEL ARCHITECTURE

- **Input Features (X):** 7 raw predictor columns expanding into 43 learned encoded dimensions.
- **Target Variable (y):** selling_price (evaluated in Indian Rupees ₹).
- **Estimator:** Scikit-Learn Pipeline bundling ColumnTransformer and LinearRegression().
- **Objective Function:** Ordinary Least Squares (OLS) minimizing Residual Sum of Squares: $RSS = \Sigma (y_i - \hat{y}_i)^2$.
- **High-Cardinality Pruning:** Dropped vehicle 'name' (1,491 titles) to avoid extreme parameter explosion.

Model Evaluation: Quantitative Metrics and Diagnostic Visualizations

Evaluated on held-out test partition of 716 unseen cars (80/20 train-test split, random_state=42).

MEAN ABSOLUTE ERROR (MAE)

₹1,79,642

Test split MAE · **Train: ₹1,62,856** (Gap: +₹16,786)

ROOT MEAN SQUARED ERROR (RMSE)

₹3,84,723

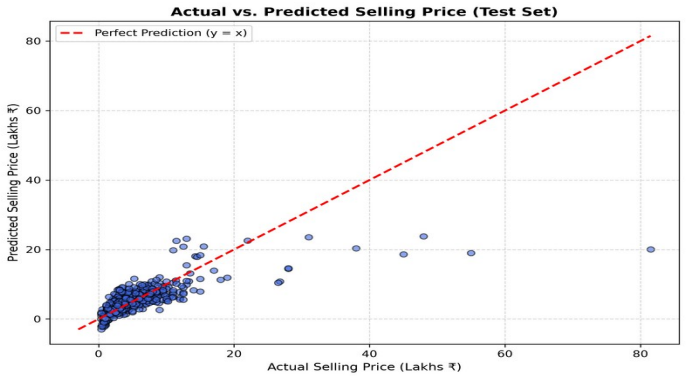
Test RMSE · **Train: ₹2,87,437** (Gap: +₹97,286)

COEFFICIENT OF DETERMINATION (R^2)

0.5405

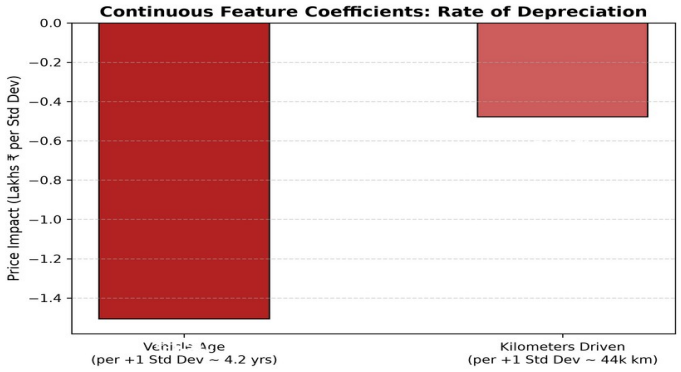
Test R^2 · **Train R^2: 0.6608** (Gap: -0.1203)

Actual vs. Predicted Price (Test Set)



Scatter aligns with red dashed line (y = x); error dispersion widens for luxury vehicles (> ₹25 Lakhs).

Depreciation Rate: Vehicle Age vs. Mileage



Standardised comparison: age penalises price 3.15× more per std dev (-₹1.51L / 4.2y) than mileage (-₹0.48L / 44k km).

Model Interpretation & Diagnostics

- **Noticeable Generalisation Gap:** R^2 drops from 0.6608 (train) to 0.5405 (test, Δ=-0.1203) and RMSE widens from ₹2.87L to ₹3.85L (+₹97.3k), indicating noticeable performance drop on unseen data.
- **Base Intercept:** ₹10,42,435.48 (model-estimated reference price at standardized mean predictor values).
- **Luxury Outlier Sensitivity:** Squared errors heavily penalize multi-million rupee luxury car residuals, elevating test RMSE.
- **Residual Distribution:** Residuals are centred approximately around zero overall, with a right-skewed tail indicating larger errors for some high-priced vehicles.

Conclusion, Limitations, Contributions & AI Disclosure

KEY EMPIRICAL FINDINGS

- **Age vs. Mileage (Standardised):** Per standard deviation of variation, vehicle age carries a 3.15× larger price penalty (-₹1,50,691 / 4.2 yrs) than odometer mileage (-₹47,754 / 44k km), reflecting normalized dataset variance rather than raw per-year vs. per-km rates.
- **Brand Equity Associations:** Relative to baseline brands, luxury European marques exhibit strong positive model associations (e.g. Land Rover +₹23.35L, Volvo +₹12.37L, Audi +₹10.28L), whereas budget marques exhibit negative relative offsets.
- **Specification Associations:** Controlling for other factors, automatic transmission (+₹3.34L over manual) and diesel fuel (+₹2.79L over petrol) are positively associated with price. These represent linear conditional associations, not guaranteed market premiums or causal effects.
- **Noticeable Generalisation Gap:** While data leakage was eliminated, evaluation reveals a noticeable performance drop on unseen test data (R^2: 0.6608 → 0.5405; RMSE: ₹2.87L → ₹3.85L), driven by luxury car dispersion.

Empirical statistical associations derived from 43 learned OLS regression coefficients.

GENUINE TECHNICAL LIMITATIONS

- **Linear Depreciation Assumption:** Ordinary Least Squares assumes constant annual depreciation, whereas real automobile valuation decays exponentially (steepest in early years 1–3).
- **Luxury Outlier Sensitivity:** Squared-error loss disproportionately penalizes multi-million rupee luxury vehicles, inflating test RMSE (₹3.85L) well above test MAE (₹1.80L).
- **Unconstrained Linear Predictions:** Standard Linear Regression does not enforce non-negativity constraints; for extreme combinations of advanced age and high mileage, the unconstrained linear hyperplane can extrapolate to unphysical negative selling prices ($\hat{y} < 0$).
- **Omitted Confounders & Non-Causality:** The dataset lacks vehicle cosmetic condition, accident history, maintenance logs, and registration city tier, precluding causal interpretations.

Highlights structural boundary constraints of linear regression on secondary automotive data.

TEAM CONTRIBUTIONS & ROLES

- **Siddharth Kumar (Execution Lead):** ML pipeline architecture, ColumnTransformer integration, model training & plotting scripts.
- **Divyaans Agarwal (Video Lead):** Project video walkthrough recording, presentation timing & submission media.
- **A Damaruk Vajra Tej (PPT Lead):** Slide architecture, narrative structure, technical synthesis, metric validation & design.
- **Nihal Ahirwar (Documentation Lead):** Comprehensive README writeup, repository structure & experimental results cataloging.

AI-Use Disclosure

Gamma: Used to create and design the presentation.

ChatGPT: Used for project-related guidance, analysis assistance and presentation review.

Claude: Used for reviewing and refining the final presentation.

Gemini 3.8 Flash: Used for formatting the initial README.md and creating the ASCII representation in main.py.

The team reviewed and verified AI-assisted content against the actual project code, dataset and model results.